Rehearse Before Execute: Digital Twin Action-Outcome Evaluation for UAV Recovery

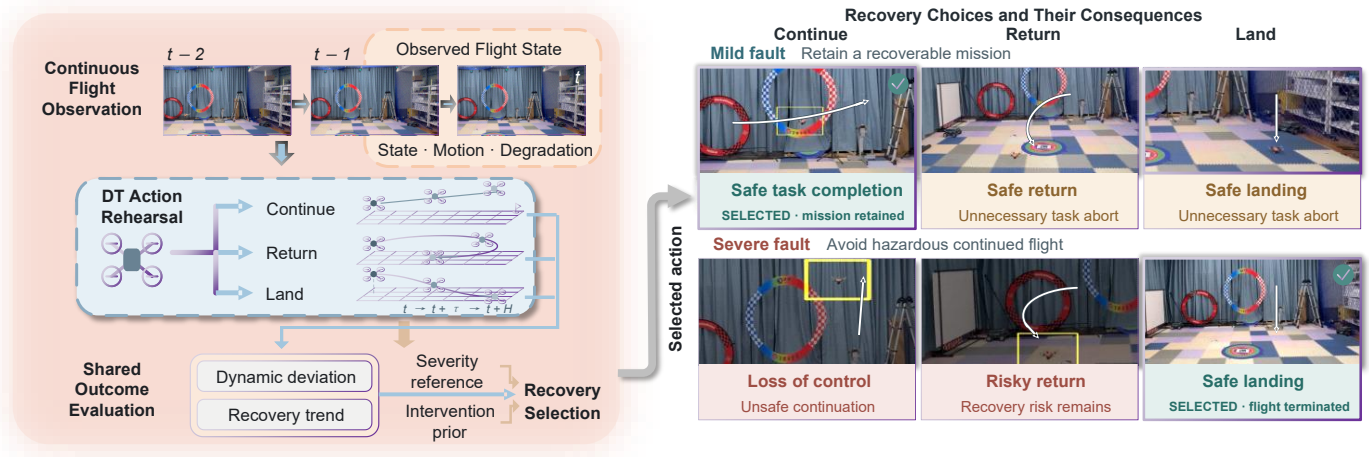


Fig. 1. Rehearse before execute: using candidate-action consequences to support UAV recovery selection. Left: continuous observations supply the current flight state; the DT rehearses Continue, Return, and Land, and shared action-outcome evaluation combines flight degradation with candidate responses to select a recovery action. Right: illustrative mild- and severe-fault situations motivate retaining a recoverable mission or terminating hazardous flight.

***Abstract*— Rehearsing recovery actions before execution allows a degraded unmanned aerial vehicle (UAV) to compare their consequences before committing to a response. This paper presents a digital twin (DT)-assisted framework that virtually executes Continue, Return, and Land through flight-control-compatible interfaces. A shared normal-dynamics evaluator scores the resulting trajectories, combining action-specific responses with flight degradation to guide recovery selection. Retrospective evaluation achieves 92.06% recovery-action agreement. Comparisons with aggressive, conservative, and threshold-based strategies demonstrate mission-retention and recovery benefits in selected simulation scenarios, complemented by physical-flight evidence of distinct action responses. The framework supports balancing mission preservation and recovery through pre-execution evaluation of candidate actions.**

***Index Terms*— Digital twin, unmanned aerial vehicle, recovery decision, safety, consequence evaluation.**

I. Introduction and Related Work

When a UAV degrades in flight, emergency decisions must balance fault-related risk against unnecessary mission interruption, choosing whether to continue, return, or land [1]. For an observable, controllable aircraft, fault severity alone is insufficient: continuation sustains mission demands, while return and landing entail distinct manoeuvres. Neither an action's label nor its apparent conservatism establishes its safety; selection requires anticipating the consequences of candidate responses before execution. *The key question is not only 'How severe is the fault?" but also 'How will the flight evolve under each response?"*

From detecting degradation to selecting a response. Safety assessment, reliability analysis, and rule-based evaluation quantify UAV degradation [2]–[5], but cannot establish whether an alternative action will mitigate or amplify it, as each response imposes distinct control demands. Recovery selection therefore requires a further capability: **Can the trajectories of unexecuted candidate actions be predicted and compared at decision time?** Such foresight complements fault severity with action-conditioned dynamics, grounding response selection in the consequences of each manoeuvre under the current degradation [6], [7].

How do existing approaches generate emergency actions? Emergency actions are commonly selected through predefined contingency logic or learned policies. Automated contingency management organizes predefined procedures [8], state machines encode decisions as explicit states and transitions [9], and PX4 failsafes trigger modes such as Return and Land under specified conditions [10]. These mechanisms offer transparent, traceable responses but rely on established *condition-action* mappings. Reinforcement learning instead learns policies through interaction [11], [12], incorporating safety constraints, physical information, or hybrid action representations [13]–[15]. Yet even when trained in a digital twin (DT) [16], a policy that directly outputs an action need not predict each candidate's consequences at decision time. Both paradigms thus select without explicitly rehearsing alternatives under the current degradation. This *selection without rehearsal* enables rapid response but leaves the dynamic consequences of unselected actions unexamined.

Rehearse candidate actions in the DT before selection.

Virtual execution can ground recovery selection in consequences specific to the current degradation. Existing DT research [17] supports UAV state representation, fault detection, and dynamics modelling [18], while flight-control-compatible toolchains enable virtual aircraft to receive recovery commands and generate responses [19]. These capabilities motivate a rehearsal-based approach in which the DT reproduces the degraded condition, simulates alternative responses, and informs action selection through their predicted consequences. This pre-execution comparison is central to the DT's value in emergency decision-making. The physical aircraft can execute only the selected action, whereas virtual flight can reveal the evolving consequences of multiple alternatives. Such situation-specific evidence complements the experience encoded in rules and policies, enabling recovery through rehearsal before selection.

Why rehearse before emergency selection? Emergency response presents a practical tension. Assessing an action's consequences requires execution, yet a degraded aircraft cannot safely test every alternative. Predefined responses also leave their situation-specific effects unexamined. DT rehearsal offers a compromise by retaining existing procedures while transferring trial execution to a virtual aircraft, allowing selection to account for predicted dynamics under current degradation. Applied to emergency response, rehearsal evaluates whether continued flight sustains deviations or a recovery manoeuvre steers the aircraft toward nominal dynamics. Mission continuation can therefore be assessed against comparable predicted responses rather than inferred from an action's label or presumed conservatism.

Can virtual responses represent real flight? Simulation credibility requires model verification, data validation, and comparison with operational responses [20], [21], while sim-to-real research highlights simulation's predictive relevance to real-world performance [22], [23]. We argue that fidelity for emergency rehearsal lies in preserving the dynamic features that determine recovery outcomes. The physical-virtual comparisons in Section III-B show that the DT produces distributions closer to real-flight records for several recovery-relevant variables. Combined with the selection results in Section III-C, these findings establish an initial foundation for using DT rehearsal to inform emergency decisions through action-conditioned evidence in the examined scenarios. Further response calibration and broader scenario coverage are needed to strengthen its role in risk assessment and autonomous recovery under complex degradation.

Building on this perspective, we propose a DT-assisted framework that rehearses candidate recovery commands and compares their responses through shared normal-dynamics evaluation (Fig. 1). The illustration contrasts mission retention under mild degradation with flight termination under severe degradation, motivating consequence-aware intervention. Experiments connect response fidelity, selection mechanisms, and physical candidate execution. The main contributions are as follows.

i) ***Candidate-action rehearsal architecture.*** Existing recovery commands are integrated into a flight-control-compatible DT to generate response evidence before physical execution.
i) ***Shared normal-dynamics evaluation.*** Nominal flight behaviour provides a common reference for quantifying dynamic deviations and recovery trends across candidate responses.
i) ***Recovery selection from flight degradation and action outcomes.*** The framework combines a flight degradation score with candidate-response costs to rank recovery actions, accounting for both degradation and the response to each action.

II. Methods

A. Recovery-selection problem

Consider a UAV experiencing in-mission degradation while retaining state feedback and configured-command execution. At the mission-management level, recovery selection compares candidate responses under current conditions. Let x_t, y_t, and m_t denote UAV state, degradation observation, and mission context (waypoints and recovery configuration). The candidate set is

$$\Pi = (\texttt{Continue}, \texttt{Return}, \texttt{Land}). \tag{1}$$

Remark 1 (Emergency-action semantics). `Continue`, `Return`, and `Land` retain the waypoint mission, execute the configured return procedure, and initiate local landing, respectively. These modes reflect automated contingency management and flight-controller failsafes [8], [10]. The selector chooses the high-level mode; the flight controller executes it through continuous feedback.

Each candidate π is rehearsed from a shared DT initial state,

$$\hat{\tau}_\pi = \mathcal{F}_{\mathrm{DT}}(z^\pi_{t_0}, \pi, \Delta T), \qquad z^\pi_{t_0} = \mathcal{S}(x_{t_0}, y_{t_0}, m_{t_0}), \tag{2}$$

where $\mathcal{S}$ initializes the virtual state and $\mathcal{F}_{\mathrm{DT}}$ propagates the candidate over horizon ΔT. The shared evaluator converts each candidate trajectory with sufficient valid observations into an anomaly sequence,

$$R^\pi = \mathcal{E}_\theta(\hat{\tau}_\pi), \tag{3}$$

using feature construction, nominal-dynamics prediction, and error normalization (Section II-C). Once all candidates are evaluated, their costs combine dynamic deviation, recovery trends, and intervention preference. Selection follows

$$\pi^* = \arg\min_{\pi \in \Pi} J(\pi). \tag{4}$$

Ties are resolved in the order Continue, Return, then Land. The selector outputs the chosen mode and comparative costs for execution by the flight controller.

B. DT architecture and candidate-action rehearsal

*1) **Expanding candidate consequences from current observations**:* As shown in Fig. 2, state synchronization transfers flight state, degradation observations, and mission context to the DT. Three branches rehearse Continue, Return, and Land from a common initial state, isolating command-dependent responses under identical degradation and retaining trajectories $\hat{\tau}_C$, $\hat{\tau}_R$, and $\hat{\tau}_L$. A shared evaluator applies the same predictor

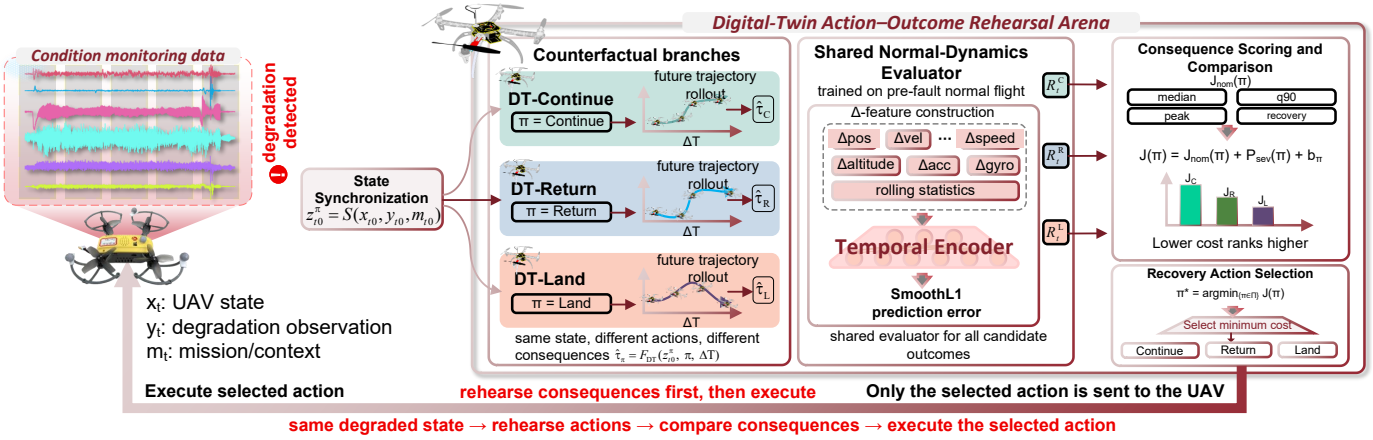
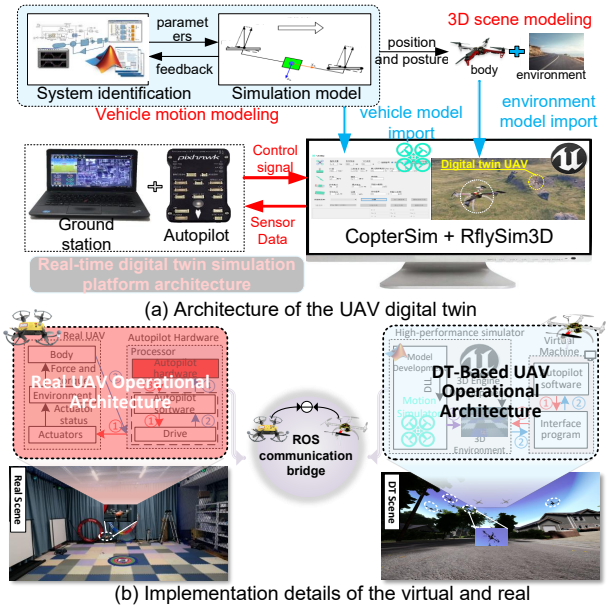


Fig. 2. Overall framework of DT-assisted action-outcome decision making. Left: current state, degradation observations, and mission context are mapped to candidate DT initial states. Center: Continue, Return, and Land generate rehearsal trajectories, which the same normal-dynamics evaluator converts into anomaly sequences. Right: candidate-response costs, the flight degradation score, and action biases determine the total costs. The selector chooses the minimum-cost action from Continue, Return, and Land, and only that command is sent to the physical UAV.



(a) Architecture of the UAV digital twin

(b) Implementation details of the virtual and real parallel operation mechanism

Fig. 3. DT implementation and physical–virtual interaction. (a) System identification, a motion model, and a 3D scene form the virtual flight environment, coupled to the flight controller via control signals and sensor feedback. (b) A Robot Operating System (ROS) bridge links the physical UAV to the DT interface, relaying flight observations and recovery commands.

Algorithm 1: DT rehearsal, recovery selection, and command dispatch

Input: Flight state, candidates Π, horizon ΔT, frozen evaluator, centers c_π, weight λ_s, biases b_π.
Output: Action π^*, costs $\{J(\pi)\}$, and execution trace.

if *online execution* **then**
  Synchronize candidate states and check fresh telemetry
  Acquire and log the current state snapshot (t_0); verify inter-vehicle state consistency
  Start candidate responses concurrently; record t_r
  Collect responses over ΔT and compute R^π using the shared frozen evaluator
else
  Load archived trajectories and compute R^π using the same frozen evaluator
Record final-scoring start t_e
foreach $\pi \in \Pi$ **do**
  Compute $J_{\text{nom}}(\pi)$ from R^π
$S \leftarrow J_{\text{nom}}(\texttt{Continue})$
foreach $\pi \in \Pi$ **do**
  $J(\pi) \leftarrow J_{\text{nom}}(\pi) + \lambda_s |S - c_\pi| + b_\pi$
$\pi^* \leftarrow \arg\min_{\pi \in \Pi} J(\pi)$; break exact ties by Continue, Return, then Land
Record scoring completion t_s
if *online execution* **then**
  Dispatch π^* automatically; record the first matching command transmission t_{tx}
  Log the command acknowledgement when available
Save state checks, candidate responses, costs, selected action, and timestamps
return π^*, $\{J(\pi)\}$, *and the execution trace*

and nominal-error scale to all candidates. Scoring combines action-specific deviations and recovery trends with the flight degradation score to account for intervention need. Their combined costs rank the candidates, and only the selected command is executed physically.

*2) **Translating commands into flight responses**:* Fig. 3(a) shows DT construction and control interaction: system identification parameterizes the dynamics model, which, coupled with a 3D environment and feedback controller, generates closed-loop responses to candidate commands under degraded dynamics. Multicopter translational and rotational dynamics are [24]

$$
\begin{aligned}
{}^{e}\dot{\mathbf{p}} &= {}^{e}\mathbf{v}, \quad {}^{b}\dot{\mathbf{v}} = -[{}^{b}\boldsymbol{\omega}]_{\times}\,{}^{b}\mathbf{v} + g\mathbf{R}_{be}^{T}\mathbf{e}_3 + \frac{\mathbf{F}}{m}, \\
\mathbf{J}\,{}^{b}\dot{\boldsymbol{\omega}} &= -{}^{b}\boldsymbol{\omega} \times (\mathbf{J}\,{}^{b}\boldsymbol{\omega}) + \boldsymbol{\tau} + \mathbf{G}_a.
\end{aligned}
\tag{5}
$$

Superscripts e and b denote earth and body frames; $\mathbf{p}$, $\mathbf{v}$, and $\boldsymbol{\omega}$ are position, velocity, and angular rate; m and $\mathbf{J}$ are mass and inertia; $\mathbf{F}$, $\boldsymbol{\tau}$, and $\mathbf{G}_a$ are force, control torque, and additional torque. $\mathbf{R}_{be}$ maps body-frame vectors to earth coordinates, and $\mathbf{e}_3$ is the vertical unit vector. These dynamics connect control inputs to observable trajectories. Fig. 3(b) depicts the physical-virtual interface: the UAV supplies flight observations, while a ROS bridge transfers states and commands to the DT and re-

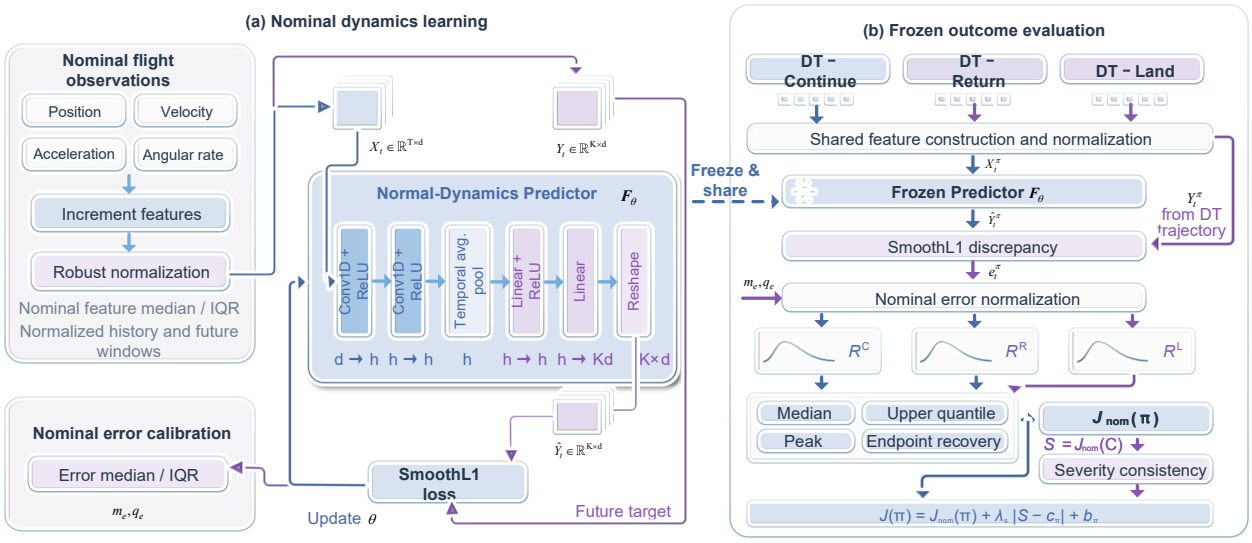


Fig. 4. Shared normal-dynamics evaluator training and application. (a) Nominal pre-fault data yield normalized increment features; temporal convolutions, average pooling, and an MLP map history $\boldsymbol{X_t}$ to $\hat{\boldsymbol{Y}}_{\boldsymbol{t}}$ for comparison with $\boldsymbol{Y_t}$, while nominal errors calibrate $\boldsymbol{m_e}$ and $\boldsymbol{q_e}$. (b) The frozen predictor evaluates Continue, Return, and Land trajectories; prediction errors form anomaly sequences whose four statistics determine candidate costs, with the Continue cost defining the flight degradation score in (12). $\boldsymbol{C}$, $\boldsymbol{R}$, and $\boldsymbol{L}$ denote actions; $\boldsymbol{T}$, $\boldsymbol{K}$, $\boldsymbol{d}$, and $\boldsymbol{h}$ denote history length, prediction horizon, feature dimension, and hidden width. Flame/snowflake icons mark trainable/frozen parameters, dashed lines indicate updates, and gray denotes reused calibration.

turns simulated responses. The RflySim-ROS-PX4 toolchain[1] enables virtual execution of high-level candidate commands through flight-control-compatible interfaces [19].

C. Normal-dynamics evaluation and recovery-action scoring

*1) **Establishing a common scale from nominal flight**:* As shown in Figs. 2 and 4, a shared predictor trained exclusively on action-unlabeled nominal flight windows is frozen and applied to all three DT candidates, converting short-horizon feature-prediction errors into anomaly sequences and candidate costs. To reduce sensitivity to absolute position and mission phase, we use incremental features:

$$\mathbf{f}_t = [\Delta\mathbf{p}_t, \Delta\mathbf{v}_t, \Delta v_t, \Delta h_t, \Delta\mathbf{a}_t, \Delta a_t, \Delta\boldsymbol{\omega}_t, \Delta\omega_t, \eta_t]. \quad (6)$$

Vector increments capture changes in position, velocity, acceleration, and angular rate; Δv_t and Δh_t denote speed and height changes, and Δa_t and $\Delta\omega_t$ the absolute changes in acceleration and angular-rate magnitudes. Auxiliary features η_t comprise the displacement-increment norm and short-window standard deviations of acceleration-increment norm, angular-rate-increment norm, and speed increment. All features are normalized using nominal medians and interquartile ranges (IQRs), with identical transformations across candidates to ensure a common evaluation scale.

*2) **Predicting short-horizon responses from feature histories**:* Given a normalized history $X_t = [\mathbf{f}_{t-T+1}, \ldots, \mathbf{f}_t]$, the predictor estimates the next K feature vectors,

$$\hat{Y}_t = F_\theta(X_t), \qquad Y_t = [\mathbf{f}_{t+1}, \ldots, \mathbf{f}_{t+K}]. \quad (7)$$

Training minimizes the nominal prediction error,

$$\mathcal{L}_{\text{nom}} = \text{SmoothL1}(Y_t, \hat{Y}_t). \quad (8)$$

[1] https://rflysim.com/doc/en/

F_θ comprises two temporal 1D convolutional layers with ReLU, adaptive average pooling, and a two-layer MLP that maps each history to $K \times d$ outputs, reshaped into $\hat{Y}_t$ (Fig. 4(a)). Adam optimizes the network using nominal segments; architecture and training settings are listed in Table I. After training, the predictor, feature normalizer, and nominal-error statistics are frozen and shared across all candidates, ensuring consistent evaluation scales (Fig. 4(b)).

*3) **Converting prediction inconsistency into anomaly sequences**:* In Fig. 4(b), each rehearsed DT trajectory undergoes the training-time feature transformation and sliding-window evaluation. The frozen predictor maps history X_t^π to future features, which are compared with the corresponding rehearsed targets Y_t^π:

$$e_t^\pi = \text{SmoothL1}\big(Y_t^\pi, F_\theta(X_t^\pi)\big). \quad (9)$$

Here, the DT generates the complete trajectory, while the predictor quantifies deviations from learned nominal dynamics. To normalize these errors against nominal prediction statistics, we use the median m_e and IQR q_e:

$$R_t^\pi = \max\left(0, \frac{e_t^\pi - m_e}{\max(q_e, \epsilon)}\right), \quad (10)$$

where ϵ ensures numerical stability. Sliding-window evaluation yields anomaly sequences R^C, R^R, and R^L (Fig. 2), expressed on a shared nominal scale for candidate comparison. The clipped values $A_t^\pi = \text{clip}(R_t^\pi, 0, A_{\max})$ serve visualization only; decision-making retains unclipped R_t^π to preserve distinctions among large deviations.

*4) **Extracting recovery consequences from anomaly sequences**:* The lower panel of Fig. 4(b) summarizes each anomaly sequence using four statistics, as shown in Fig. 2: median, upper quantile, peak, and endpoint recovery. These

jointly define the nominal recovery cost:

$$J_{\text{nom}}(\pi) = w_1 \,\text{med}(R^\pi) + w_2 Q_\rho(R^\pi) + w_3 \max(R^\pi) - w_4(R_1^\pi - R_N^\pi). \tag{11}$$

The median captures typical deviation, while Q_ρ and the peak quantify elevated and extreme anomalies. Here, Q_ρ denotes the ρ-quantile, N the number of valid samples, and w_1–w_4 the corresponding weights. The endpoint term rewards a net anomaly decrease and penalizes an increase, capturing overall recovery while the quantile and peak retain sensitivity to severe fluctuations within the rehearsal interval.

*5) **Relating dynamic consequences to intervention need:*** A low candidate-response cost can result from ending the mission early. To account for degradation while retaining the current mission, we define the flight degradation score as,

$$S = J_{\text{nom}}(\texttt{Continue}), \tag{12}$$

Here, S summarizes the Continue response using the same evaluator and cost function applied to all candidates. To relate this score to the recovery actions, we use:

$$P_{\text{sev}}(\pi) = \lambda_s |S - c_\pi|. \tag{13}$$

where c_π is the score center associated with action π and λ_s is its weight. Although S is shared, the centers differ across actions, so this term affects their relative costs. The final cost is

$$J(\pi) = J_{\text{nom}}(\pi) + P_{\text{sev}}(\pi) + b_\pi. \tag{14}$$

The three terms account for the candidate response, the distance between S and the action's score center, and a fixed action bias. Return and Land responses enter through their respective candidate-response costs. The score centers are calculated by combining fixed values with development-data medians:

$$m_\pi = \text{med}_{i \in \mathcal{C}_\pi} S_i, \qquad c_\pi = \alpha e_\pi + (1-\alpha) m_\pi. \tag{15}$$

Here, $\mathcal{C}_\pi$ contains development records grouped by the scenario-defined recovery action π, m_π is their median flight degradation score, and e_π is a fixed value used to calculate the center. The coefficient α determines the relative weights of e_π and m_π. Values are listed in Table I.

D. Candidate-action selection

Algorithm 1 integrates rehearsal, evaluation, and execution in Fig. 2. It generates the three candidate trajectories, computes their nominal costs, establishes the shared flight degradation score, and evaluates final costs before sending the minimum-cost action to the flight controller. Selection applies (14) to each complete anomaly sequence, without additional penalties from temporal means or standard deviations of candidate-cost histories. Rolling feature statistics support representation, whereas across-record variability in the results summarizes performance.

III. Experiments

Experiments address three research questions:

- **RQ1: Does the DT reproduce recorded degraded-flight responses?** Real-flight and DT traces and distributions are compared with PD-AR rollouts, which combine proportional-derivative (PD) feedback with first-order autoregressive (AR) residuals.
- **RQ2: How do rehearsed consequences inform recovery selection, and which components matter?** recovery-action agreement, intervention errors, scoring, propagation, ablations, and parameter sensitivity are evaluated.
- **RQ3: What physical responses follow candidate recovery actions?** DT cost rankings are related to recorded physical executions.

A. Experimental setup and configuration

1) Platform: Experiments use an FS-X150/PX4 UAV[2]. The ROS) connects the vehicle, motion-capture feedback, RflySim/CopterSim[3] replicas, and action dispatcher. Continue, Return, and Land share command semantics across virtual rehearsal and physical execution.

2) Scenarios and policy reference: Three accelerometer-noise levels and three trajectory geometries define nine degradation scenarios and 189 replay records. AccStd denotes the configured noise standard deviation; AccStd3, AccStd9, and AccStd15 correspond to levels 3, 9, and 15, respectively. The recovery actions follow documented operational principles: FAA guidance addresses safe operation and emergency risk reduction [25], while PX4 defines Return and Land as graded failsafe actions [26]. For retrospective evaluation, the three noise levels map to Continue, Return, and Land, representing mission retention, return, and landing-based termination. This numerical mapping is study-specific; the cited sources support the underlying principles and action semantics. All trajectories and augmented records at each noise level share a label assigned by its setting identifier, independently of S, c_π, and J. Proposed selects actions by comparing candidate costs.

Remark 3 (Action agreement and execution outcomes). Action agreement quantifies consistency with scenario-defined labels, which share the intervention ordering used to calculate score centers. Mission completion and recovery outcomes are evaluated separately using execution records in Section III-C.5.

3) Comparison methods: Six controls assess alternative recovery rules. *Reactive* maps the Continue anomaly peak to actions using fixed thresholds; *Rule* uses median and upper-quantile thresholds. *Fixed Return* and *Fixed Land* always select their named actions. *DT clipped anomaly* clips anomalies before arbitration, whereas *DT normality only* removes severity consistency and fixed action biases. Two additional controls isolate scoring and propagation effects. *DT quadratic scoring* retains the DT candidates, flight degradation score, centers, and biases but replaces the direct nominal cost with a scale-aligned quadratic feature-deviation cost. *PD-AR rollout* replaces post-fault propagation with PD position/velocity

[2] http://en.feisilab.com/?product_31/62.html
[3] https://rflysim.com/doc/en/3.Soft/coptersim.html

TABLE I
PARAMETER CONFIGURATION AND BASIS

Configuration	Basis
$(d, T, K) = (20, 50, 10)$; width 32; conv/MLP depth 2/2	Retained architecture
Adam; learning rate 10^{-3}; weight decay 10^{-4}	Retained checkpoint
$(m_e, q_e) = (0.177, 0.060)$; $\epsilon = 10^{-6}$	Nominal statistics; numerical floor
$\mathbf{w} = (0.40, 0.30, 0.10, 0.20)$; $\rho = 0.90$; $\lambda_s = 1$	Balances characteristic deviations, upper-tail anomalies, peaks, and recovery trends
$\mathbf{b} = (0, 0.15, 0.50)$	Bias rises with task interruption; prefer minimal intervention at comparable costs
$\mathbf{e} = (140, 300, 600)$; $\mathbf{m} = (133.556, 396.601, 652.898)$	Fixed values; development-data medians
$\alpha = 0.8$; $\mathbf{c} = (138.711, 319.320, 610.580)$	Weighted calculation of score centers
$A_{\max} = 100$; $\kappa = 0.318$	Clipping bound; scale alignment
AccStd: 3, 9, 15; A15-R Continue payload (real/DT): 45/15	Recorded scenarios
Horizon/step/gap: $20/0.1/0.5$ s; 20 stochastic runs	Unified time-domain and sampling-grid comparison

Note. Vectors follow Continue, Return, Land; $\mathbf{w} = (w_1, w_2, w_3, w_4)$. Each m_π is the median original Continue score at level π, excluding augmented records; $c_\pi = \alpha e_\pi + (1 - \alpha) m_\pi$. Nominal errors define $m_e = \mathrm{med}(e_t)$ and $q_e = \mathrm{IQR}(e_t)$. The scale $\kappa = \mathrm{med}(|J_{\mathrm{nom}}|)/\mathrm{med}(G)$, where G is mean squared standardized feature deviation.

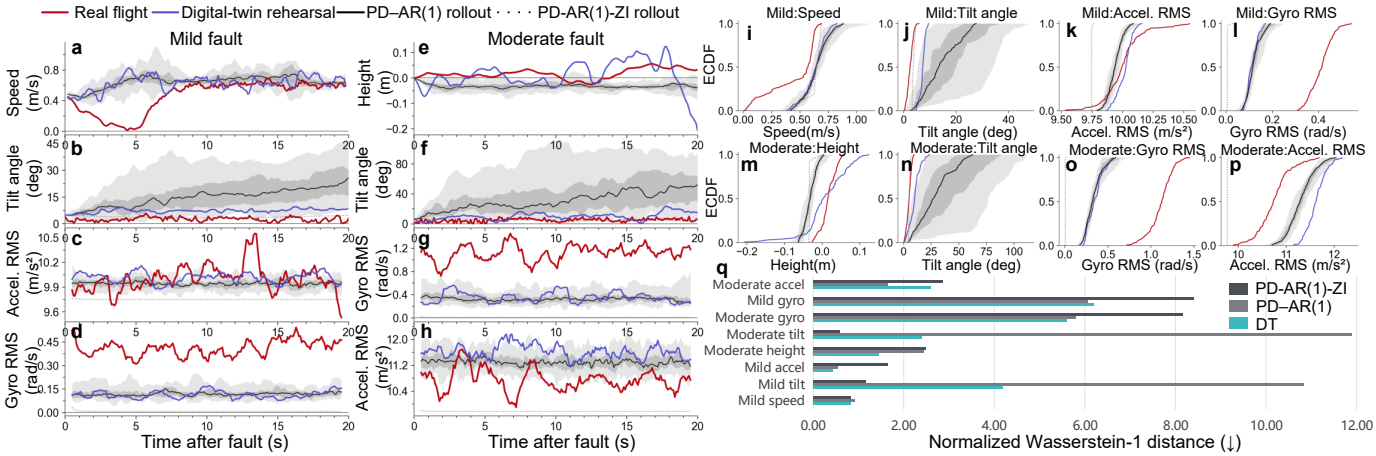


Fig. 5. Flight and distribution comparisons of digital-twin rehearsal and PD-AR rollouts against real flight records (PD: proportional–derivative control; AR(1): first-order autoregressive model). (a–h) Responses under mild/moderate faults: (a–d) mild fault metrics (horizontal speed, tilt angle, acceleration RMS, angular-rate RMS); (e–h) moderate fault metrics (relative height, tilt angle, angular-rate RMS, acceleration RMS). (i–p) Empirical cumulative distribution functions (ECDFs) for distribution matching. Stochastic curves show medians across runs; dark/light bands indicate interquartile and min–max ranges. (q) Normalized Wasserstein-1 distances (lower = closer distributional agreement); stochastic-model bars show median distances across runs.

TABLE II
QUANTITATIVE PERFORMANCE COMPARISON OF DECISION METHODS

Method	Decision Quality				Error Rates		Metric Definitions
	Saf. ↑	Corr. ↑	Sev. ↑	ME ↑	FA ↓	MD ↓	↑: higher is better; ↓: lower is better.
Reactive	0.5557	35.45	0.6772	34.18	93.65	**0.00**	**Saf.** (internal, 0–1): DT-cost agreement.
Rule	0.8424	66.67	0.8333	**73.33**	**0.00**	50.00	**Corr.** (Correctness, %): recovery-action agreement.
Fixed Return	0.6995	33.33	0.6667	60.00	100.00	50.00	**Sev.** (Severity, 0–1): severity-level match.
Fixed Land	0.3570	33.33	0.5000	20.00	100.00	**0.00**	**ME** (proxy, %): fixed action preference.
DT clipped anomaly	0.3243	13.76	0.3651	67.62	93.65	73.81	**FA** (False Abort, %): intervention when GT is Continue.
DT normality only	0.5182	32.80	0.5238	40.11	87.30	24.60	**MD** (Missed Danger, %): intervention below the GT level.
DT quadratic scoring	0.9737	91.01	0.9497	62.33	**0.00**	10.32	Same DT candidates; quadratic direct cost.
PD-AR rollout	0.8701	69.84	0.8492	72.06	**0.00**	45.24	Alternative propagation; same evaluator and J.
DT(Proposed)	**1.0000**	**92.06**	**0.9603**	63.17	**0.00**	11.90	Saf. uses updated costs for every method.

Metric definitions. Let GT denote the expert policy reference, with reference action g_i for record i, selected action a_i, candidate set Π_i, and final candidate cost $J_i(\pi)$ from Eq. (14). • ***Saf.*** $= \frac{1}{N}\sum_i \left[1 - \frac{J_i(a_i) - \min_{\pi\in\Pi_i} J_i(\pi)}{\max_{\pi\in\Pi_i} J_i(\pi) - \min_{\pi\in\Pi_i} J_i(\pi) + \epsilon}\right]$. • ***Corr.*** $= 100N^{-1}\sum_i \mathbb{1}[a_i = g_i]$. • ***Sev.*** $= N^{-1}\sum_i \left(1 - \frac{|L(a_i) - L(g_i)|}{2}\right)$, where $L(\cdot)$ maps Continue/Return/Land to 0/1/2. • ***ME*** $= 100N^{-1}\sum_i M(a_i)$, with $M(\mathtt{Continue}, \mathtt{Return}, \mathtt{Land}) = (1, 0.6, 0.2)$. • ***FA*** $= 100|\mathcal{D}_C|^{-1}\sum_{g_i = C} \mathbb{1}[a_i \neq C]$. • ***MD*** $= 100|\mathcal{D}_{\neg C}|^{-1}\sum_{g_i \neq C} \mathbb{1}[L(a_i) < L(g_i)]$.

feedback and first-order autoregressive residuals, retaining the frozen evaluator and full selection criterion.

B. Flight-response fidelity (RQ1)

*1) **Evaluation objective**:* DT rehearsal requires virtual responses to preserve recovery-relevant dynamics of degraded flight. RQ1 evaluates translational, attitude, and inertial responses using the time traces, empirical cumulative distribution functions (ECDFs), and normalized Wasserstein-1 distances in Fig. 5; lower distances indicate closer distributional agreement with real-flight records. The figure compares paired DT and real-flight records for Circle under mild faults and Figure8 under moderate faults during post-fault Continue, with PD-AR and its zero-innovation (ZI) variant as controls.

*2) **Response comparison**:* Fig. 5 shows closer agreement between DT and real-flight distributions than stochastic PD-

TABLE III
SCENARIO-LEVEL RETROSPECTIVE ACTION-OUTCOME REPLAY AND BASELINE COMPARISON

Scenario Setting		Decision Result			DT Rehearsal Evidence			
Scenario	GT	Reactive	Rule	**Proposed**	S	$J_C \downarrow$	$J_R \downarrow$	$J_L \downarrow$
VeryLight	`Continue`	✗	✓	✓	120.6 ± 12.4	139.2	337.6	596.0
Light	`Continue`	✗	✓	✓	100.5 ± 18.1	138.7	322.0	603.7
Moderate	`Return`	✗	✓	✓	354.6 ± 41.0	570.4	356.3	515.6
Serious	`Return`	✗	✓	✓	321.9 ± 33.9	505.1	307.5	567.1
Critical	`Land`	✓	✗	✓	496.7 ± 69.6	854.7	658.6	553.3
Borderline	`Continue`	✗	✓	✓	112.9 ± 16.0	139.6	342.3	586.1
TrajectoryShift	`Return`	✗	✓	✓	318.8 ± 41.9	499.0	323.2	540.6
Severe-Rect	`Land`	✓	✗	✓	501.9 ± 65.3	865.2	724.0	641.1
HighManeuver	`Land`	✓	✗	✓	743.4 ± 95.5	1348.1	1111.7	549.4

Scenario and metric definitions. The left color bar denotes ordered scenario severity, with darker shades indicating more severe cases. Scenarios span mild recoverable degradation (VeryLight, Light, Borderline), `Return`-level recoverable degradation (Moderate, Serious, TrajectoryShift), and `Land`-level severe degradation (Critical, Severe-Rect, HighManeuver). ✓/✗denote agreement/disagreement with the scenario-defined recovery action (Section III-A). S is the flight degradation score defined in Eq. (12), reported as mean ± standard deviation. J_C, J_R, and J_L are the final candidate costs of `Continue`, `Return`, and `Land` from Eq. (14); lower is better, and the minimum determines the DT-selected action.

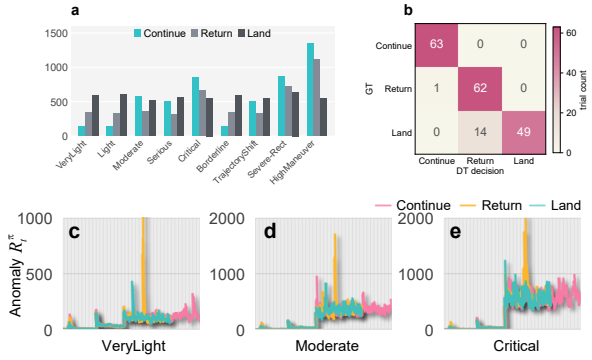


Fig. 6. Retrospective action-outcome replay evidence. (a) Mean candidate costs across scenarios. (b) Confusion matrix comparing scenario-defined recovery actions with DT-selected actions.. (c)–(e) Raw anomaly trajectories $\boldsymbol{R_t^\pi}$ in representative VeryLight, Moderate, and Critical scenarios.

TABLE IV
ABLATION RESULTS FOR KEY CONSEQUENCE-EVALUATION COMPONENTS

Changed Component	Decision Quality			Error Rates	
Variant	Corr. ↑	Sev. ↑	ME ↑	FA ↓	MD ↓
Proposed	**92.06**	**0.9603**	63.17	**0.00**	**11.90**
Remove P_{sev}	32.80	0.5238	40.11	87.30	24.60
Remove $\text{med}(R^\pi)$ in J_{nom}	88.89	0.9444	64.44	**0.00**	16.67
Remove $\max(R^\pi)$ in J_{nom}	91.01	0.9550	63.60	**0.00**	13.49
Remove $\max(R^{\text{C}})$ in S	69.84	0.8439	72.49	**0.00**	45.24
Remove $\max(R^\pi)$ in both	66.14	0.8307	73.54	**0.00**	50.79
Clip R_t^{C} for S	35.45	0.5450	**96.40**	**0.00**	96.83

Note. Symbols follow Sec. II-C: R_t^π in (10), J_{nom} in (11), S in (12), and P_{sev} in (13); R^{C} denotes Continue. Rows labeled J_{nom} alter only direct candidate costs, keeping S fixed; 'rows labeled S alter only the degradation-score calculation. "Both" removes the peak from both paths. The clipping bound is specified in Table I. Centers, other weights, and biases remain fixed; changes to S may alter its scale relative to the fixed centers. Metric definitions follow Table II; bold marks the column optimum.

AR for several recovery-relevant variables. Under mild faults, normalized distances decrease from 0.910 to 0.817 for horizontal speed and from 0.550 to 0.433 for RMS acceleration. Under moderate faults, the relative-height distance falls from 2.437 to 1.452. Tilt distances decrease from 10.818 to 4.194 under mild faults and from 11.876 to 2.393 under moderate faults. These results indicate improved agreement across translational, attitude, and selected inertial responses. Overall, the DT more closely reproduces real-flight response distributions than stochastic PD-AR in the examined scenarios, supporting its use as an empirical basis for rehearsal. However, fidelity gains vary across variables and controls. RQ2 therefore examines whether these response representations translate into improved recovery selection.

C. Recovery-action selection and component contributions (RQ2)

*1) **Evaluation objective**:* RQ2 evaluates recovery-action agreement and identifies the contributions of unified scoring. Aggregate metrics quantify overall performance, scenario-level rankings reveal decision evidence, and propagation controls, ablations, and sensitivity analyses examine the factors driving performance.

*2) **Overall decision comparison**:* Table II shows that Proposed achieves the highest recovery-action agreement (92.06%) and severity match (0.9603) among nine methods, with zero false aborts. Reactive yields zero MD but 93.65% FA, indicating excessive intervention; Rule yields zero FA but 50.00% MD, indicating insufficient intervention. Fixed-action baselines cannot adapt to degraded responses and match only one of the three action-label categories. Proposed more frequently selects the recovery action assigned to each scenario. Its Saf. value of 1.0000 follows from minimizing the same cost used by this metric. Mission completion and recovery outcomes after action execution are examined separately in Section III-C.5.

*3) **Scenario-level consequence ranking**:* Table III reports the flight degradation score S and final action costs. Fig. 6 links mean costs and representative anomaly trajectories to the confusion matrix. The action rankings reflect both candidate-response costs and the distance between the flight degradation score and each action's center. Most mismatches occur between adjacent actions, particularly Return selections in scenarios labeled Land. Mean costs summarize the rankings, while the confusion matrix shows record-level differences.

*4) **Propagation and scoring controls**:* We replace either post-fault propagation or direct scoring while retaining the remaining components to isolate their contributions. Table II reports recovery-action agreement of 92.06%, 91.01%, and 69.84% for Proposed, DT quadratic scoring, and PD-AR rollout, respectively. Replacing direct scoring changes agreement by 1.05 percentage points, whereas replacing propagation

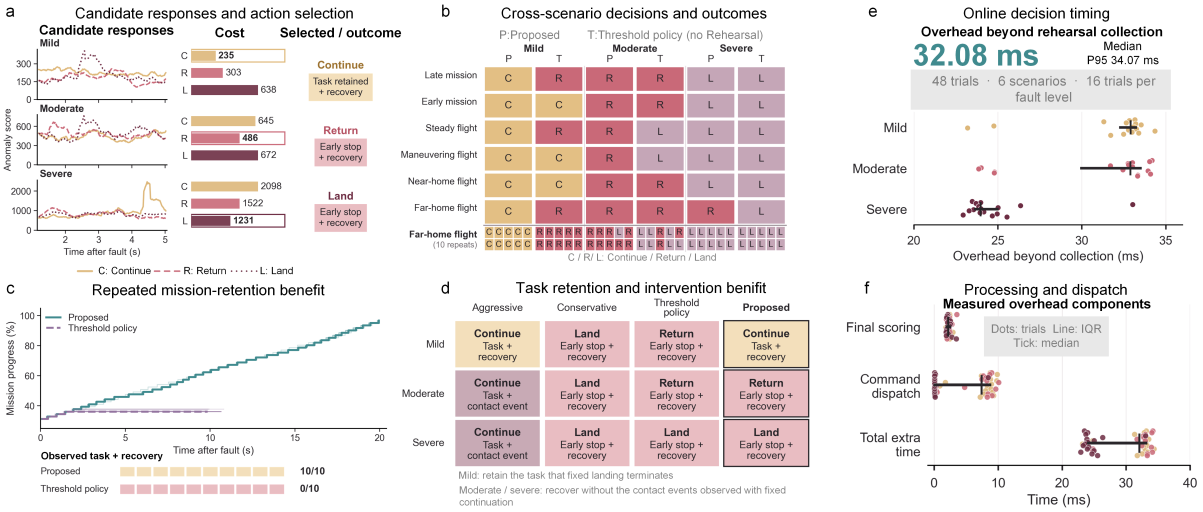


Fig. 7. Candidate-response scoring, mission retention, and recovery outcomes in closed-loop simulation. (a) Responses, composite costs, and selected actions under three fault levels in the late-mission scenario. (b) Decisions and outcomes across six scenarios. (c) Mission progress under mild degradation far from home, expressed as the percentage of planned checkpoints reached. (d) Outcomes under aggressive (Continue), conservative (Land), threshold, and Proposed policies. (e) Computational overhead. (f) Final scoring, command dispatch, and total additional latency.

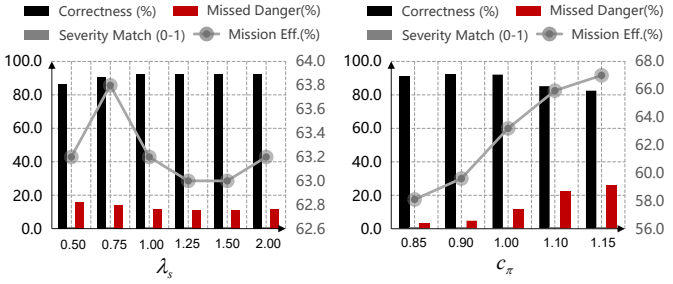


Fig. 8. Parameter sensitivity of the decision layer. Left: effect of the severity-consistency weight $\boldsymbol{\lambda_s}$ in Eq. (13). Right: uniform scaling of the centers $\boldsymbol{c_\pi}$ in Table I. Severity match retains its 0–1 scale; other metrics are percentages.



Fig. 9. Candidate rehearsal and mission continuation under mild faults (A3-C). The upper panel shows acceleration-based degradation observations, the center compares virtual Land, Return, and Continue trajectories, and the right shows physical mission continuation. Table V confirms that Continue has the lowest cost and longest flight duration, whereas Return and Land divert from the original mission.

changes it by 22.22 percentage points. Propagation affects both the flight degradation score and candidate-response costs, so this comparison evaluates their combined effect rather than the separate contribution of Return and Land rehearsal.

5) ***Closed-loop simulation: mission retention and recovery outcomes****:* Recovery selection was evaluated in RflySim/CopterSim–PX4 simulations using three simulated vehicles: the execution vehicle supplies the Continue response, while two companion vehicles rehearse Return and Land. The evaluation spans mission stage, flight motion, and home-relative location, with 102 executions across three fault levels (Fig. 7). In the far-home scenario under mild faults, Proposed completed the mission with controlled recovery in all ten trials, whereas the threshold policy returned prematurely (Fig. 7(c)). Proposed achieved 96.7% mission progress, versus 36.1%–37.7% for the threshold policy, improving mission retention without compromising the observed recovery outcome.

Late-mission cases link candidate-response scoring to intervention choices (Fig. 7(a),(d)). Proposed selected Continue, Return, and Land for mild, moderate, and severe faults, respectively, preserving the mission under mild degradation and achieving controlled recovery under moderate and severe faults. Aggressive continuation caused touchdown-limit exceedances, while threshold and conservative policies also achieved controlled recovery in these cases. These results illustrate selective intervention rather than uniformly early termination. The broader scenario matrix summarizes action selection (Fig. 7(b)). For online execution, host timestamps link state checks, candidate scoring, and command transmission following Algorithm 1. Across 48 Proposed executions, the median overhead beyond the fixed 5 s rehearsal-collection window was 32.08 ms (p95: 34.07 ms; Fig. 7(e),(f)). Candidate responses are collected within this window and evaluated online.

6) ***Component ablations****:* Table IV separately modifies candidate-response costs and the flight degradation score, with centers, biases, trajectories, and labels fixed.

Removing severity consistency P_{sev} reduces recovery-action agreement from 92.06% to 32.80% and raises false aborts to 87.30%, indicating that dynamic smoothness alone cannot capture intervention needs. Aligning the flight degradation score with intervention levels is thus critical to mission retention and action agreement. With the score fixed, removing the median or peak from direct costs lowers agreement to 88.89% and 91.01%, respectively, while increasing missed-

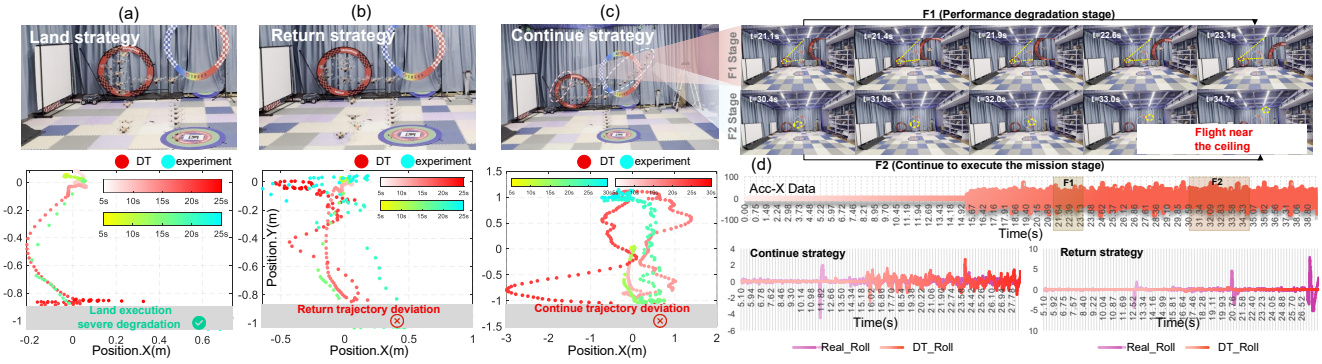


Fig. 10. Candidate-action responses under severe faults (A15-R). (a)–(c) Physical and DT horizontal trajectories for Land, Return, and Continue; color gradients indicate recorded time. (d) Real-flight frames and acceleration changes during degradation and continued mission execution, together with physical–DT roll responses under Continue and Return. Ceiling proximity during continued flight and differences among candidate trajectories provide context for timely flight termination. Check/cross symbols indicate agreement/disagreement with the scenario-defined recovery action. These trajectories and signals describe action consequences; they do not establish a strictly paired fidelity ranking or independently certify safety.

TABLE V
DT CANDIDATE COSTS AND DESCRIPTIVE REAL-FLIGHT MEASUREMENTS

Scenario	GT	DT	S	$J_C \downarrow$	$J_R \downarrow$	$J_L \downarrow$	$T_C/T_R/T_L$ (s)	$D^{xy}_{C/R/L}$ (m)	$a^{\max}_{C/R/L}$ (m/s^2)	e_{DR} (m)
A3-C	`Continue`	✓`Continue`	114.8	138.7	349.1	592.9	32.2/13.2/10.1	3.51/1.64/1.66	5.3/26.3/3.9	2.12
A9-F8	`Return`	✓`Return`	350.6	562.5	244.3	483.3	32.1/13.4/10.3	3.38/1.82/1.80	20.8/21.5/9.1	1.84
A15-R	`Land`	✓`Land`	472.3	805.8	536.4	520.3	24.5/13.5/10.3	1.89/0.98/0.97	54.7/46.7/44.5	1.49

Scenario and metric definitions. A3-C, A9-F8, and A15-R denote representative mild, moderate, and severe fault cases, corresponding to the VeryLight, TrajectoryShift, and Severe-Rect scenarios in Table III, respectively. GT denotes the scenario-defined recovery action; ✓indicates action agreement. T, D^{xy}, and $a^{\max}$ represent the recorded flight duration, maximum horizontal displacement, and peak acceleration for `Continue`/`Return`/`Land`, respectively. e_{DR} is the retained historical DT–real position RMSE for the selected action, not an error from strictly matched initial states and event times.

danger rates. The median captures persistent deviations and contributes more to direct ranking, whereas the peak penalizes extreme excursions. The peak has greater influence in the reference pathway: removing it from S reduces agreement to 69.84%, and removing it from both pathways lowers agreement to 66.14%. Clipping anomalies only in S further reduces agreement to 35.45% and raises missed danger to 96.83%. These results reflect the role of the Continue peak in defining the shared severity reference. Removing or truncating large-magnitude anomalies depresses S relative to the fixed centers, disrupting their calibration consistency. Reference magnitudes must therefore remain consistent with center-point calibration.

*7) **Parameter sensitivity**:* With the normal-dynamics model fixed, Fig. 8 examines recovery-action agreement under variations in severity weight λ_s and uniform scaling of centers c_π, isolating scoring and calibration effects.

As reported in Table I, agreement remains 86.24%–92.59% across the tested λ_s range; the default achieves 92.06% agreement with zero false aborts, indicating stability to weight variation. Center scaling more directly shifts intervention tendency: missed-danger rates range from 3.17% to 26.19% across the extreme tested scales. The results indicate relative stability over the tested weight range, while the score centers must remain consistent with the scale of S. Overall, RQ2 indicates that the Continue reference primarily establishes intervention need, whereas candidate costs distinguish action-specific dynamic responses. Their complementary roles jointly determine recovery selection.

D. Real-flight validation of candidate-action execution (RQ3)

A3-C, A9-F8, and A15-R correspond to VeryLight, TrajectoryShift, and Severe-Rect in Table III, with rehearsal and real-flight records for all three actions. Table V reports candidate costs, flight durations, maximum horizontal displacements, and peak accelerations. Fig. 9 and 10 show mild- and severe-fault executions to explain the motivations for mission continuation and recovery intervention.

Mild faults: retaining the mission under degradation. In A3-C, Continue has the lowest cost (138.7), followed by Return (349.1) and Land (592.9). Its longer duration (32.2 s vs. 13.2 and 10.1 s) and larger horizontal range distinguish mission continuation from recovery. However, Return reaches higher peak acceleration (26.3 m/s^2) than Continue (5.3 m/s^2) or Land (3.9 m/s^2), showing that conservative intervention need not minimize every dynamic quantity. Consistent with Fig. 9 and RQ2, selection combines the Continue reference with full candidate responses; neither degradation alone nor minimum peak acceleration dictates intervention. Longer flight supports mission-retention interpretation but does not establish safe completion.

Moderate faults: balancing response dynamics and intervention need. In A9-F8, Return has the lowest cost (244.3), versus 562.5 for Continue and 483.3 for Land. Physical Return records show shorter duration and smaller maximum horizontal displacement than Continue, yet similar peak acceleration (21.5 vs. 20.8 m/s^2); Land reaches 9.1 m/s^2. Thus, selection cannot be explained by a single peak or duration. Return instead balances action-specific response dynamics with the

intervention preference encoded by the Continue reference. Table V quantifies the physical differences underlying this ranking.

Severe faults: limiting continued-flight consequences. In A15-R, Land has the lowest cost (520.3), closely followed by Return (536.4), while Continue is substantially higher (805.8). Physical Continue lasts 24.5 s and reaches 54.7 m/s^2 peak acceleration; Return lasts 13.5 s at 46.7 m/s^2, and Land 10.3 s at 44.5 m/s^2. Fig. 10 shows pronounced motion changes and ceiling proximity during continued flight, while physical and DT trajectories and roll responses differ under Return and Continue. These observations motivate prioritizing flight termination under severe degradation, but also show that mode switching does not eliminate anomalies. The physical and DT fault payloads differ in this case (Table I); these records therefore illustrate candidate behavior, not the safety benefit of Proposed over alternatives.

IV. Conclusion and Limitations

This study investigates DT-based consequence evaluation for UAV emergency actions. The framework combines a flight degradation score with candidate-response costs to compare recovery actions under degraded conditions.. Real-flight cases show that conservative interventions can incur substantial transient costs, motivating joint assessment of continued-flight degradation and recovery consequences to support interpretable trade-offs between mission retention and timely intervention.

As an initial feasibility study, experiments cover three trajectories, graded accelerometer noise, and indoor execution of Continue, Return, and Land. Actuator and compound faults, complex disturbances, and broader operating conditions remain untested; the method also assumes available state feedback and candidate-command execution. Future work will extend fault types and outdoor missions, while addressing the real-time constraints of state synchronization, parallel rehearsal, scoring, and command activation. Computational acceleration and adaptive rehearsal horizons will be investigated to quantify trade-offs among prediction fidelity, computational cost, and decision latency in closed-loop flight.

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